

EXPERIMENT 6

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 6 – Implement VLAN and VLAN Trunking Protocols.

Aim: To implement Virtual Local Area Networks (VLANs) and VLAN trunking on network switches, and to configure communication between different VLANs using appropriate trunking protocols.

Objective

- To understand the concept and purpose of VLANs in a computer network.
- To create and configure multiple VLANs on a network switch.
- To assign switch ports to specific VLANs.
- To configure trunk links for carrying traffic from multiple VLANs.
- To understand the operation of VLAN trunking protocols such as IEEE 802.1Q.
- To verify connectivity and VLAN communication using suitable network commands.

Theory:

L2

Switch:

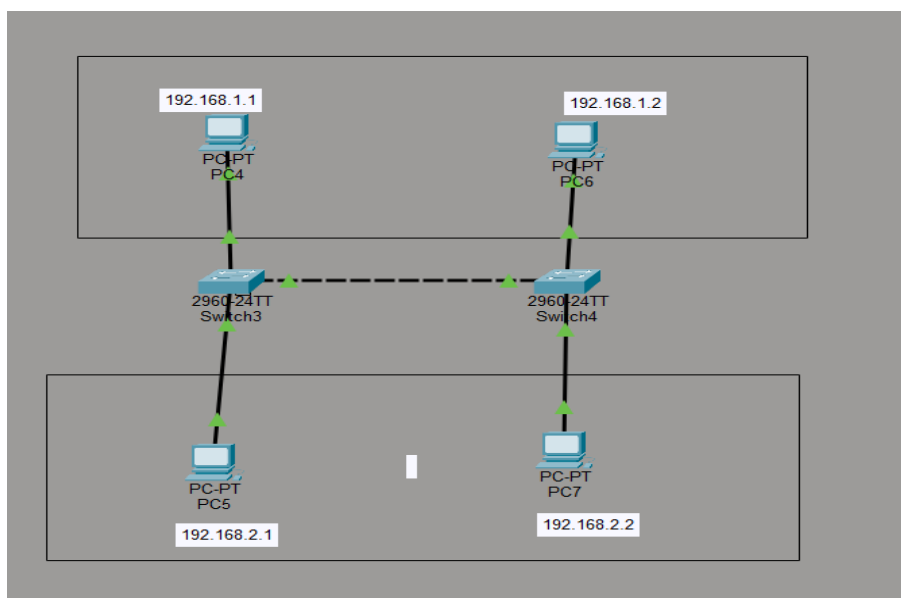
A Layer 2 switch operates at the **Data Link Layer (Layer 2)** of the OSI model and forwards Ethernet frames using **MAC addresses**. It learns the MAC addresses of connected devices and stores them in a MAC address table to determine the appropriate port for forwarding frames. L2 switches support **VLANs**, allowing a physical network to be divided into multiple logical broadcast domains. They can be configured with **access ports** for individual VLANs and **trunk ports** to carry traffic from multiple VLANs. However, an L2 switch cannot normally perform routing or communication between different VLANs.

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L3 Switch:

A Layer 3 switch operates at both the **Data Link Layer (Layer 2)** and **Network Layer (Layer 3)** of the OSI model. It performs normal switching using MAC addresses and can also route packets between different networks or VLANs using **IP addresses** and routing tables. L3 switches support **Inter-VLAN Routing**, usually through Switch Virtual Interfaces (SVIs), allowing devices belonging to different VLANs to communicate without requiring a separate router. They are commonly used in larger networks where both high-speed switching and routing capabilities are required.

Practical Steps:



Step 1: Take two switches 2960 Cisco and take 4 PC's.

Step 2: Use Copper-Straight Through cables to connect them.

PC1 → SW1 Fa0/1

PC2 → SW1 Fa0/2

PC3 → SW2 Fa0/1

PC4 → SW2 Fa0/2

Then connect two switches:

SW1 Gi0/1 → SW2 Gi0/1

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Step 3:

Assign IP Addresses:

PC1:

IP Address: 192.168.10.11

Subnet Mask: 255.255.255.0

PC3:

IP Address: 192.168.10.12

Subnet Mask: 255.255.255.0

PC1 and PC3 are going to belong to: VLAN 10

So their IP addresses are also from: 192.168.10.0

PC2:

IP Address: 192.168.20.11

Subnet Mask: 255.255.255.0

PC4:

IP Address: 192.168.20.12

Subnet Mask: 255.255.255.0

VLAN 20.

Step 4: Configuring VLAN on SW1 (Switch 1).

SW1 -> GO TO CLI.

Switch>

Enter Privileged Mode:

Type: Enable

Switch#

Enter Configuration Mode:

Type: Configure Terminal or Conf t

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Switch(config)#

Step 5: Create VLAN 10 on SW1

Type: VLAN 10

Switch(config-vlan)#

Give it a name: name STUDENTS

Now VLAN 10 has been created.

VLAN 10 -> STUDENTS.

Step 6: Create VLAN 20 on SW1

Type: Exit

Then: VLAN 20

And

Name FACULTY

Now SW1 has:

VLAN 10 -> STUDENTS

VLAN 20 -> FACULTY

Step 7: Assigning PORTS to VLAN.

Creating a VLAN does **not** automatically put PCs into that VLAN.

We still have to tell the switch: “Fa0/1 belongs to VLAN 10”

And “Fa0/1 belongs to VLAN 20”

Step 8: Configure Fa0/1 as an Access Port.

Type: Exit

Now enter the interface: interface fa0/1

You should see: Switch(config-if)#

Now tell the switch that this is an access port:

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Type: Switchport mode access

Then assign it to VLAN 10: switchport access vlan 10

So the complete configuration will be:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

By this we tell SW1 that fa0/1 -> Access Port -> VLAN 10.

Therefore, PC1- > Fa0/1 -> VLAN 10.

Step 9: Configure Fa0/2 for VLAN 20:

Type: Exit

```
interface fa0/2
```

```
then
```

```
switchport mode access
```

```
then
```

```
switchport access vlan 10.
```

Step 10: Repeat the same process for SW2.

Step 11: Now we have to configure the TRUNK.

This is the most important part of the experiment.

We have: SW1 Gi0/1 ===== Gi0/1 SW2

Currently, this is just a normal switch-to-switch connection.

We need to tell both switches:

"This link is going to carry multiple VLANs."

Therefore, we configure it as a **trunk**.

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Step 12:

On SW1:

enable

configure terminal

then

interface gigabitEthernet 0/1

then

switchport mode trunk

Now Gig0/1 is now the trunk port.

Step 13: Repeat the same process for SW2.

Learning Outcome

After completing this experiment, the learner will be able to:

1. Understand the basic concepts and working principles of **Layer 2 and Layer 3 switches**.
2. Configure and manage **VLANs** on network switches.
3. Configure **trunk links** to carry traffic between multiple VLANs.
4. Implement and verify **Inter-VLAN Routing** using a Layer 3 switch.